

RetailEdge Inc.

Final AWS Architecture

Three-Tier Migration

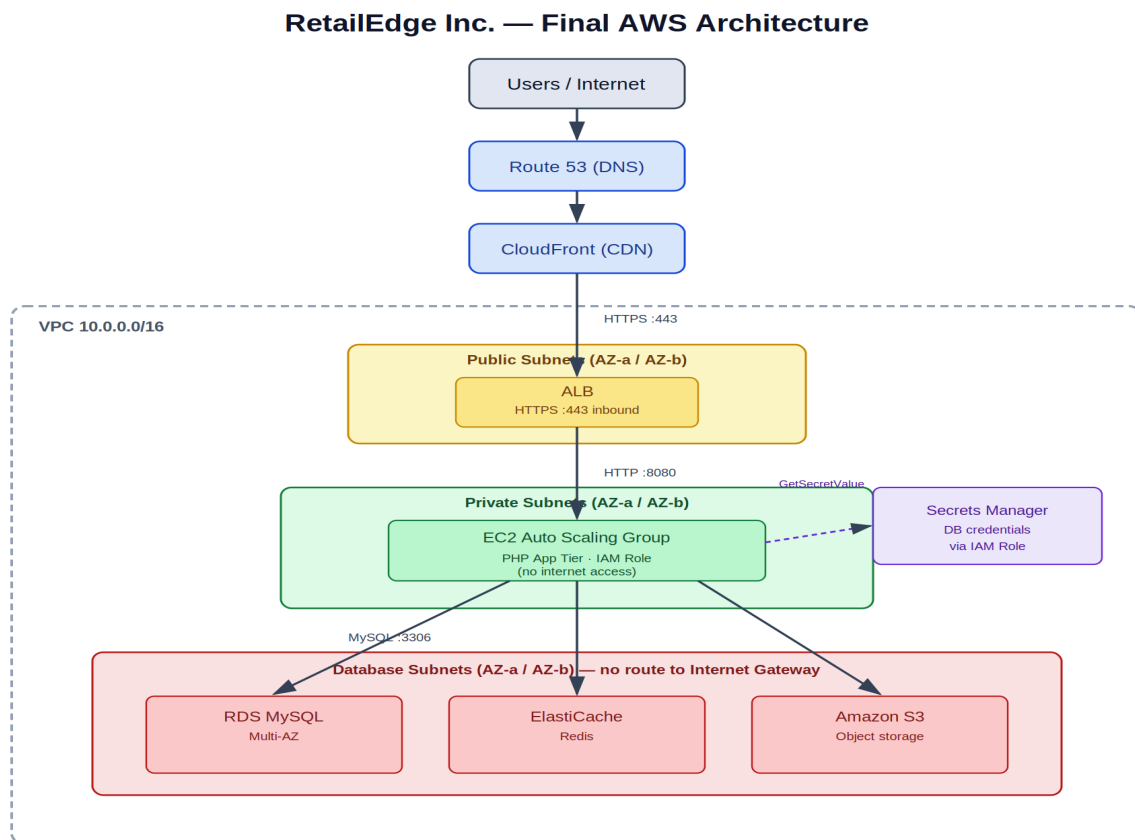
This document describes the final approved AWS architecture for migrating RetailEdge Inc.'s e-commerce platform from three bare-metal servers to a highly available, three-tier architecture on AWS.

1. Architecture Overview

The solution is organized into three main layers:

- Web Tier — DNS, content delivery, and the public entry point
- Application Tier — Auto-scaled EC2 instances running the PHP application
- Data Tier — Managed database, caching, and object storage services

2. High-Level Diagram



Detailed Architecture [here](#)

3. Web Tier

- Route 53 provides DNS resolution for the application domain.
- CloudFront serves as the content delivery layer, caching static content close to users.
- An Application Load Balancer (ALB) is the primary public entry point and accepts HTTPS traffic (port 443) from the internet.

4. Application Tier

- The ALB distributes traffic across an EC2 Auto Scaling Group hosting the PHP application.
- Application instances sit in private subnets and are not directly reachable from the internet.
- Auto Scaling adjusts capacity automatically to handle demand spikes such as Black Friday.
- Each instance uses a least-privilege IAM role to reach AWS services — no AWS credentials are stored on the instance.

5. Data Tier

- Amazon RDS for MySQL (Multi-AZ) is the primary relational database, improving availability.
- ElastiCache for Redis provides application-level caching.
- Amazon S3 provides object and file storage.
- AWS Secrets Manager securely stores database credentials; the app tier retrieves them at runtime via its IAM role — never hard-coded in application code, AMIs, or User Data.
- Database resources are isolated from direct internet access and accept only the required application traffic.

6. Network Architecture

A single VPC (10.0.0.0/16) spans two Availability Zones for segmentation and high availability, with three subnet layers:

- Public subnets: internet-facing resources such as the ALB
- Private subnets: the EC2 application tier
- Database subnets: the RDS database tier (no route to the Internet Gateway)

Subnet Layout

Layer	Availability Zone A	Availability Zone B
Public	10.0.1.0/24	10.0.2.0/24
Private	10.0.11.0/24	10.0.12.0/24
Database	10.0.21.0/24	10.0.22.0/24

7. Security Architecture

Security follows the principle of least privilege: resources may only communicate over the ports and with the components they explicitly require.

Security Group Rules

Security Group	Port	Allowed Source	Purpose
ALB SG	443 (HTTPS)	Internet (0.0.0.0/0)	Public entry point
Application SG	8080 (HTTP)	ALB SG only	App tier traffic
RDS SG	3306 (MySQL)	Application SG only	Database access

- The application tier uses an IAM role to access AWS services without long-lived AWS credentials on the EC2 instances.

- Database subnets have no route to the Internet Gateway, adding a further layer of isolation.
- Security Groups provide stateful, resource-level control; Network ACLs can add subnet-level control where required.

Request Flow

Step	From	To	Protocol / Port
1	Internet	ALB	HTTPS : 443
2	ALB	EC2 Application Tier	HTTP : 8080
3	EC2 Application Tier	RDS MySQL	MySQL : 3306

Database Credential Flow

Step	From	To	Action
1	EC2 Application Tier	AWS Secrets Manager	IAM Role: GetSecretValue
2	AWS Secrets Manager	EC2 Application Tier	Returns DB credentials
3	EC2 Application Tier	RDS MySQL	Authenticated connection

8. End-to-End Request Flow (Summary)

- Users → Route 53 → CloudFront → Application Load Balancer
- Application Load Balancer → EC2 Auto Scaling Group
- EC2 Auto Scaling Group → Data Services:
 - RDS MySQL (Multi-AZ)
 - ElastiCache for Redis
 - Amazon S3

Database credentials are managed separately through AWS Secrets Manager and retrieved by the application tier when required.

9. Design Scope

This document represents the final high-level and network design.

The following implementation-level details are defined during the detailed design and implementation phases:

- Route tables and NAT requirements
- Network ACL rules
- Availability Zone placement of individual resources
- Instance sizing and Auto Scaling policies
- Storage configuration
- Secrets Manager integration details
- IAM policies
- Monitoring and logging
- Backup policies